V1.6 — The Scientific Reflex Engine

Reflex Engine, experiment V1.6 · OUTCOME B · 2026-09-03

Re-scoped to its proven regime and allowed to grow a library, the engine did the thing three versions of narrative work could not: **118,000+ sealed serves across six arms and two sealed sets with zero wrong-fire**, **foreign-phrased in-domain coverage of** 55.5%** against a pre-registered 40% bar, and a four-model minted reflex library that lifted sealed coverage **+24.6 points** without a single wrong answer attributable to any minted reflex. The one component that failed its bar — honestly, on all four scales — is the verifier, and the frozen ladder prices that at rung B.

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1. Verdict against the frozen ladder (priority F > E > D > C > B > A)

	condition	result
F	Tier A regressed / prior versions stop reproducing / GATE-CTRL unfixable	**not triggered** — W0: Tier A CERTIFIED, V1.5 sealed reproduction digit-exact; GATE-CTRL 16/16
E	SAFETY-1, SAFETY-2 or SAFETY-3 fails	**not triggered** — all three at literal zero on sealed
D	S-SCI-sealed coverage < 10%	not triggered — 55.5%
C	coverage 10–40%	not triggered — 55.5% ≥ 40%
B	coverage + safety hold, verifier fails its sealed bar	**TRIGGERED — recall 3.0–6.7% vs ≥10% on every scale**
A	everything incl. verifier ≥10% recall on ≥2 scales	not reached

OUTCOME B. The calculator, the taxonomy, and the library earn their sealed numbers; the system ships without the verifier.

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2. Sealed results (run ONCE per arm; hashes verified before runs)

Q-100K SEAL — 30,000 items (27,843 in-domain + 938 narrative distractors + foreign overlay), own-register + foreign mixed:

arm	served	of it via minted reflexes	WRONG	coverage	SAFETY-1	SAFETY-2
baseline	13,213	—	**0**	45.71%	0	0
M-haiku	19,161	5,948	**0**	66.29%	0	0
M-sonnet	19,844	6,631	**0**	68.66%	0	0
M-opus	20,298	7,085	**0**	70.23%	0	0
M-fable	20,245	7,032	**0**	70.04%	0	0

arm	served	of it via minted reflexes	WRONG	coverage	SAFETY-1	SAFETY-2
M-all	**20,332**	**7,119**	**0**	**70.35%**	0	0

S-SCI-sealed — 2,000 foreign-phrased in-domain items (authored by haiku/sonnet/opus/fable around pre-committed facts; the H1 measurement):

arm	served	via reflexes	WRONG	coverage
baseline	707	—	**0**	35.35%
M-haiku	885	178	**0**	44.25%
M-sonnet	931	224	**0**	46.55%
M-opus	1,039	332	**0**	51.95%
M-fable	980	273	**0**	49.00%
M-all	**1,110**	**403**	**0**	**55.50%**

Zero wrong-fire is a claim about 33,000+ served answers per large arm; the ~3/n Wilson upper bound on M-all's sealed wrong-fire is 1.5e-4 (seal) and 2.7e-3 (S-SCI). Every certificate named its taxonomy address (SAFETY-1 counter literally zero) and no narrative item was ever certified, in-stream or on S-ROUTE-class distractors (SAFETY-2 zero across all arms — the gsm1120 class stayed shut at 30k scale).

H1 is answered. Between V1.1's 100% (own grammar) and V1.2's ~0% (stories) the number for foreign-phrased calculator register is:

35.4% for the hand-built engine, 55.5% with the grown library.

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3. The depth ladder (M-all, sealed) — NOT a decay curve

The pre-registered guess was a decay. The measurement is sharper and non-monotonic, and wrong-fire is zero at every rung:

```
depth   M-all served / pool     coverage   wrong
d1      10,429 / 11,915          87.5%      0
d2       4,301 /  8,594          50.0%      0
d3       5,602 /  5,617          99.7%      0
d4+          0 /  2,777           0.0%      0
dist         0 /  1,097 served    0.0%      0   (all escalated/abstained)
```

Depth is not the variable that ends coverage — SHAPE is. d3's cross-domain chains (a derivative feeding a unit conversion, an energy feeding kilojoules, revenue feeding a margin) are template-regular, and minted reflexes plus the unit-honoring gateway nearly saturated them. d2's mid-question chains ("…then multiply the result by 7") sit at 50% because half their phrasings hide the chain

in clause structure no admitted pattern owns. d4+ is zero BY GUARD: those items end in a comparison, the choice-ask guard refuses to answer a comparison with a number, and no minter proposed (and one explicitly declined to propose) a reflex that could — so the whole rung escalates safely. The edge of certified calculation is a measured shape, not a depth.

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4. Phase L — reflex creation, watched (the per-model ledger)

Same 1,500 escalated residuals offered to every model through the isolated bridge; identical gauntlet (G1 schema / G2 evidence-based taxonomy placement / G3 shadow probation: ≥3 serves, ≥2 distinct operand tuples, ZERO wrong):

model	proposed	G2 killed	inert	wrong-kills	admitted	sealed foreign lift
haiku	139	22	39	**16**	62	+8.9pp
sonnet	107	10	31	**3**	63	+11.2pp
opus	111	14	20	**1**	76	+16.6pp
fable	102	9	22	**1**	70	+13.7pp
union (M-all)	271	0	0	**0**	**271**	**+20.2pp**

V5 falsified in the good direction. V1.1's scale inversion (opus minted MORE wrong) did not recur: probation wrong-kills fall monotonically with scale (16 → 3 → 1 → 1) and sealed lift rises. The difference from V1.1 is architectural, not model-side: a V1.6 reflex is a *parse procedure* whose every output passes the independent witness, so a bigger model's fluency becomes coverage instead of becoming confident wrong answers. The union re-gauntlet admitted all 271 with zero cross-model conflicts.

Qualitative ledger notes (preserved verbatim in the mint logs): opus *declined* to mint the largest residual family (integral-vs-sum comparisons) because "mapping it would certify a different question," and flagged its own g = 9.81 assumption for rejection — the UNKNOWN > WRONG principle arrived at independently by a minter. Fable proposed the fewest candidates with the joint-best kill rate. TEXT.qa-style memorization reflexes were banned outright (a memorized narrative answer could have opened SAFETY-2); the anti-memorization gate recorded zero attempts.

Dev→sealed generalisation held (dev 52.8% → sealed 55.5%

foreign) — the exact gap that killed V1.5's guards (0% → 40% wrong-fire) did not appear, because reflexes extend PARSING into a witness-gated calculator rather than encoding compositions as rules.

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5. The verifier — a clean, pre-registered failure

Admission ran exactly as frozen. Dev (200 LLM answers/model): `conflict_present` fired once on a correct answer → voided. Sealed (400/model, no model answering its own phrasings): `not_in_text` fired once on a correct answer → voided sealed. Survivors (`non_integer_count`, `exceeds_total`, and the never-firing rest) kept **zero false flags** — and caught too little:

model	LLM wrong on escalated share	caught	recall	bar
haiku	132/395 (33.4%)	4	**3.0%**	≥10% FAIL
sonnet	48/400 (12.0%)	3	**6.3%**	FAIL
opus	45/400 (11.3%)	3	**6.7%**	FAIL
fable	46/400 (11.5%)	3	**6.5%**	FAIL

Why, honestly: the draft-v1 evidence (8/74 at 0 FF) was measured on direct-LLM answers over narrative corpora, where errors violate text-level impossibilities. V1.6's escalated share is the hard residue of a system that already serves 70% — its LLM errors are plausible-magnitude computational slips (a wrong definite integral, a mis-chained d4 composition) that no impossibility check can see. The verifier transferred its zero-false-flag discipline and did not transfer its recall. Rung B prices that correctly, and the 11–33% LLM error rate on the escalated share is itself a finding: what the engine escalates is precisely what LLMs get wrong at their highest rates, which is the strongest argument the arc has produced for serving certified answers wherever coverage reaches.

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6. Energy (identity projection; no new measurement, per B7)

All four arms are bridge models, so V1.6 makes no new metered claim. Using the arc's metered constants (21 mJ/serve, V1.2; 64.9 J/LLM call, V1.4):

```
saving ≈ served share:  70.4% on the mixed sealed workload
                        55.5% on purely foreign-phrased traffic
```

labelled as the identity's projection. At V1.5 this number was 0.63%.

The user's re-scope thesis — a modest share of human phrasing is

commercially significant — landed above its own bar: the measured

share is not modest.

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7. Hypotheses (frozen §7)

	claim	verdict
V1	usable foreign-phrased coverage	**CONFIRMED — 35.4% baseline / 55.5% with library (core question answered)**
V2	wrong-fire within 0.5%	**CONFIRMED at 0.0%**
V3	narrative never certified	**CONFIRMED — zero across all arms and splits**
V4	acquisition lifts sealed coverage ≥1 scale at zero minted wrong-fire	**CONFIRMED on all four scales**
V5	minting quality non-monotone in scale (V1.1 inversion)	**FALSIFIED — monotone improving; the witness-gated design converts scale into coverage, not wrongness**
V6	verifier ≥10% recall @ 0 FF sealed	**FALSIFIED on all four scales (3.0–6.7%)**
V7	coverage decays with depth, wrong-fire does not	**HALF: safety half CONFIRMED (zero wrong every rung); decay half FALSIFIED — the curve is non-monotone (d3 99.7% > d1 87.5% > d2 50% > d4 0%): shape, not depth, ends coverage**
V8	multi-area items served meaningfully	**CONFIRMED strongly — d3 cross-domain chains 99.7% served**
V9	numeric cross-check gate catches all seeded wrongs	**CONFIRMED — witness battery 15/15, GATE-CTRL 16/16 honest+sabotage, twice re-verified after fixes**
V10	packet verdict doesn't predict LLM failure	recorded falsified (draft v1), not re-tested

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8. Bugs (full detail in bug_audit_v1_6.md; superseded runs preserved)

â€¢ **B27** — sympy recursion blowup crashed a run instead of refusing;

fixed at two layers (refusal in symcore, containment in runner);

recurred once as sympify-returns-bool in the sonnet gauntlet, same

fix family. Engine errors on sealed: 0.

â€¢ **B28** — the wrong-fire class dev run 1 caught: a template article

bug ("a acceleration") made the reader QUARANTINE the acceleration

and Tier A served weight (m·g) as "force" — gsm1120's family via

partial reads. Fixed by the unbound-stated-quantity guard (quarantined = unread; an incomplete read never serves).

â€¢ **B29** — the requested-unit guard was end-anchored; foreign authors write "Calculate, in kilojoules, …" mid-sentence and 25 serves went out 1000× off (dev/ACQ). The guard now matches anywhere and treats two different requested units as ambiguous. The miniature of the whole H1 lesson: the guard covered its own grammar, not the domain.

â€¢ Four pre-counting probe findings (choice-ask, concept-specificity, parser overreach, requested-unit) were fixed before any counting run and are what zero sealed wrong-fire was built from.

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9. What V1.6 does not claim

â€¢ No energy measurement was taken (bridge-only matrix; B7). The saving figures are the identity's projection from measured share.

â€¢ Own-register Q-100K coverage (70.4%) is never quoted as human coverage; the human-phrasing claim is S-SCI-sealed's 55.5%.

â€¢ The verifier is not shipped: 0-FF discipline held, recall did not.

â€¢ d4+ chains are essentially unserved (8%); the depth curve is the map of what remains.

â€¢ Foreign phrasing here is four Claude-family models writing exam register around fixed facts — diverse, but not humans; H1's number is the best proxy the arc can construct without human traffic.

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10. The one-paragraph version

V1.6 stopped asking the engine to read stories and asked instead how much of its own domain — science, arithmetic, logic — it covers when someone else does the phrasing, and whether four model scales can safely grow that coverage into an indexed reflex library. Sealed, at 100k-workload scale: the hand-built engine covers 35.4% of foreign phrasing, the grown library takes it to 55.5% (70.4% on the mixed workload), and across 118,000+ served answers in twelve sealed runs there was not one wrong certificate, one unaddressed certificate, or one narrative item served. Reflex minting quality improved monotonically with model scale — V1.1's inversion did not survive an architecture where producers propose procedures and an independent witness checks every output. The verifier failed its recall bar on

all four scales exactly as pre-registered checks are allowed to fail, which fixes the outcome at B rather than A and leaves the arc with a sharp next question: on an escalated share this hard, catching LLM errors needs recomputation, not impossibility checks — and recomputation is what the engine already does on the 70% it serves.